PONGTENTIOMETRY: Potentiometric Pong

Paulina Arizpe Romo, Guido De Santis, Kai Herman, Eduardo Najera

Team Members

Team Members, left to right: Eduardo Najera, Guido De Santis, Paulina Arizpe Romo, Kai Herman

Features

- **Sensing:** Arduino will detect voltage output from potentiometer and compare to ball positioning to determine win condition
- When win is sensed, piezo buzzer provides audio response by playing a victory song
- **Actuation:** Mini Servo motors will be used to provide physical feedback to moving the paddles on the screen
- **Computation:** Digital control of paddles through servo motors, as well as digital display featuring game mechanics virtual ball
- **Mechanical Design:** Custom made gears fitting the servo motors to provide optical feedback as to the direction of the pad

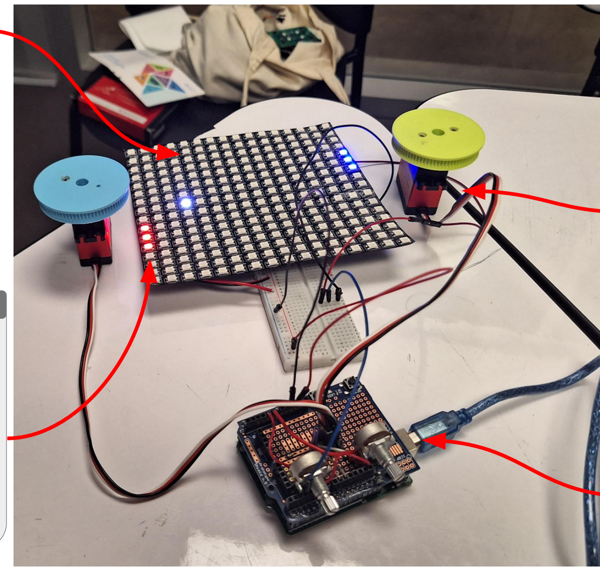
Design

Neopixel Display

- Pixel grid which allows for individual pixel control
- Simple digital display which will allow for the ball to bounce around for the players to hit

Paddles

- 2 paddles to allow for 2-player game mechanics
- Located on each side of the display and controlled by potentiometer
- Placed on the two outermost column of neopixel grid and tracked by servo motor which provides a physical response (resistance) to moving them back and forth



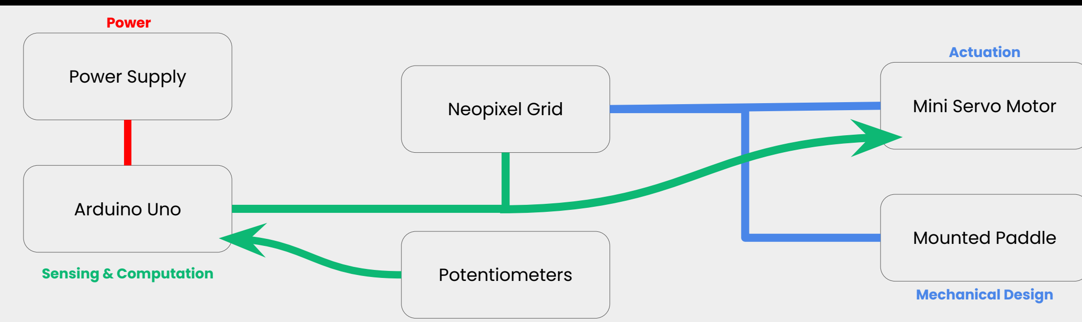
Servo Motor

- Rotary Movement
  - Nozzles are rotated 180 deg to cover all four walls.
  - Rotary movement driven by a high torque stepper motor connected to timing belt.
- Linear Movement
  - Linear motion helps cover the clean area from top to bottom.
  - Linear movement driven by a stepper motor with belt (3d printer style).

Potentiometers

- 2 Single-Turn Potentiometers allowing user interaction
- Voltage output controls movement of paddle

Block Diagram



- Arduino Uno powered via cable connection
- Arduino Uno controls Neo Pixel grid and Mini servo motor
- Mini-Servo controls paddle motion via potentiometer output